

Revisiting Implicit Curriculum in RLVR

Reproduction and local analysis of synthetic state tracking

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Abstract

Mixed training can bootstrap learning on harder tasks even when final accuracy remains well below one. We revisit the synthetic state-tracking experiment of Huang et al. [1] through an independent implementation. In the main run, mixing task lengths raises L45 terminal success from 1.03% to 11.65%. More hard tasks become solvable, although our runs do not recover the near-perfect accuracy reported at every length. We designed local measurements to examine this learning process. Attention shifts toward intended operations and local execution improves before terminal success rises appreciably. Reward-gradient norms peak in length order and then decline unevenly. Later in the main run, L15 ascent directions help L45 more often even as the L15 gradient weakens. Operation-selection accuracy then stabilizes while attention sharpens and probe outcomes polarize, indicating saturation within a partially learned policy. Other seeds reach different accuracies and retain different gradient strengths. These findings characterize hard-task bootstrapping through internal progress and cross-length reward geometry; the origin of the resulting learning states remains open.

1 Learning within a fixed task mixture

Huang et al. [1] show that a fixed mixture of task lengths can generate a learning relay from short to long tasks, reaching near-perfect terminal accuracy. We study how mixed training bootstraps harder tasks in an independent implementation. Section 2 compares terminal learning under fixed and mixed training. Sections 3 and 4 examine two local views of that process: progress within generated trajectories and the reward gradients linking task lengths.

Each prompt contains an initial state and L distinct additions modulo 96. Thus, L counts the operations in the intended solution. A fixed hidden permutation links query identifiers to operation positions. At each step, attention forms a weighted operation mixture; a frozen MLP combines it with the generated current state to predict the next state. Reward tests the final state:

$$s_L^* = (s_0 + \sum_{k=1}^L g_k) \bmod 96, \quad r = \mathbf{1}\{s_L = s_L^*\}.$$

Here s_L^* is the intended final state and s_L is the generated state. RL trains only the 64×64 attention matrix Q , initialized to zero. The shared frozen MLP solves every atomic state-operation pair. A uniform terminal guess succeeds with probability $1/96$, or about 1.04%. Longer tasks require more successive operations before this terminal feedback arrives.

Condition	Training lengths	Performance evaluation
Fixed	45 only	$L = 45$
Mixed	Uniform per sample: 5, 15, 45	$L = 5, 15, 45$

Each run uses 40,000 updates and batches of 512 trajectories. We evaluate 15,360 prompts per length every 50 updates. The main analysis follows the original run, seed 101. Two additional seeds examine variation in hard-task learning; a five-level mixture adds intermediate lengths while retaining L45. These comparisons and implementation details appear in the appendices.

2 Terminal learning under fixed and mixed training

The original $R = 3$ experiment shows learning progressing from L5 through L15 to L45, with terminal accuracies approaching one [1, Section 7.1, Figure 4a]. We reconstruct the L5/15/45 mixture and compare it with L45-only training. Both conditions share the transition network, initialization, optimizer, and update count. Figure 1 examines whether mixing lengths enables hard-task learning and how the achieved accuracies compare.

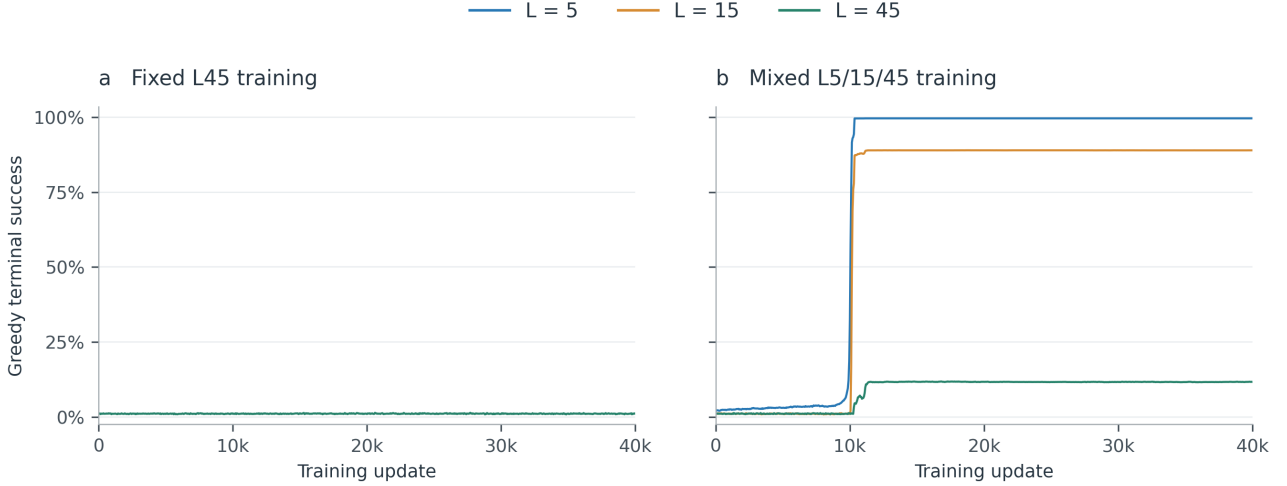


Figure 1. Terminal learning under fixed and mixed training. Left: L45-only training. Right: uniform mixed training on L5, L15, and L45. Colors denote evaluation lengths. Both panels show greedy terminal success on identical axes, retaining all 801 evaluations.

Bootstrapping more hard tasks. Fixed L45 training remains near chance. In the mixture, L5 and L15 first exceed 50% terminal success at updates 10,050 and 10,200, followed by a rise in L45 performance. L45 reaches 11.65%, compared with 1.03% in the fixed control. Mixed training thus enables success on more long tasks. The difference from the original result lies in the final accuracy: the reproduced curves do not approach one at every length.

Condition	Greedy terminal	Sampled terminal
Fixed, L = 45	1.03 ± 0.08	0.94 ± 0.08
Mixed, L = 5	99.67 ± 0.05	99.45 ± 0.06
Mixed, L = 15	88.99 ± 0.25	88.35 ± 0.26
Mixed, L = 45	11.65 ± 0.26	11.46 ± 0.26

Table 1. Final terminal success at update 40,000. Percentages with one standard error over 15,360 prompts. Greedy and sampled decoding yield similar outcomes.

Variation across runs. With the same implementation and frozen head, mixed L45 success reaches 66.17% in seed 102 but remains at 1.01% in seed 103 (Appendix C). The added five-level mixture reaches 36.88% in seed 101 (Appendix E). Mixed training can therefore produce much larger hard-task gains, with substantial variation across seeds and mixtures.

Looking inside the learning process. The terminal curves establish the hard-task gains but leave their development largely hidden. Does attention find the intended operations before success improves? Does learning on an easier length open useful directions for L45? We examine the original seed-101 run first within its generated trajectories, then through its reward gradients.

3 Internal progress before terminal success

We designed two measurements to inspect the L45 computation. *Correct attention mass* is the weight assigned to the intended operation, averaged over all steps and prompts. *Local correct-transition probability* is the probability of correctly applying that operation from the model's actual sampled current state, which may differ from the canonical state; it is also averaged over all steps and prompts. Sampled terminal success provides the outcome reference.

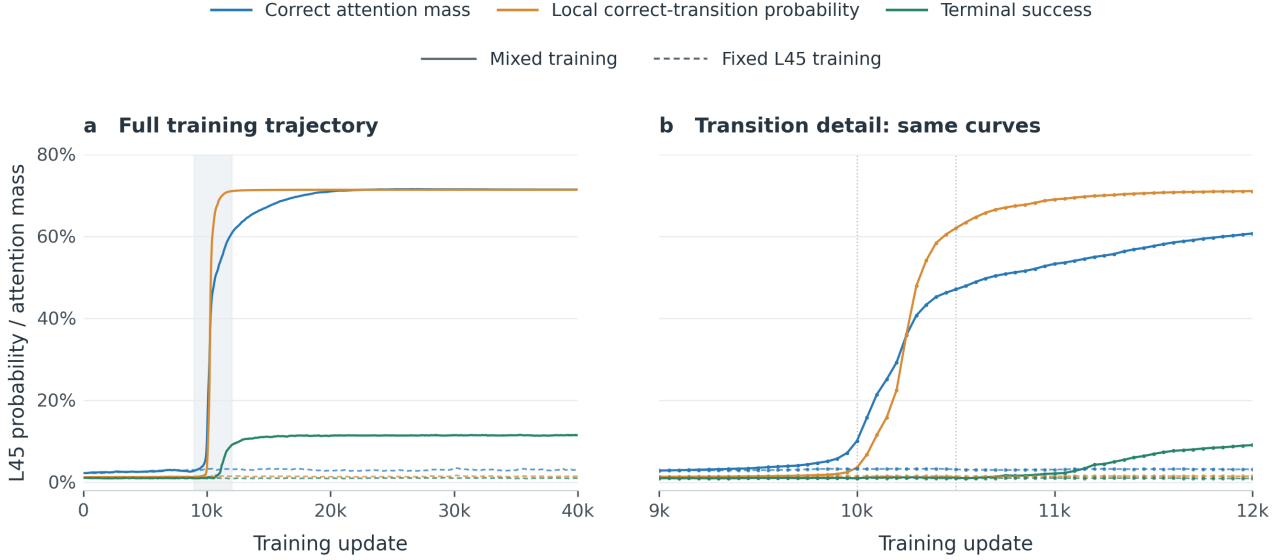


Figure 2. Two local measurements along the learning process. Both panels show the same L45 quantities and training conditions. Left: all 801 evaluations, with the transition window shaded. Right: the same curves over updates 9,000-12,000, retaining all 61 measurements. Colors identify metrics; solid and dashed lines identify mixed and fixed training. The window was selected after the run; guides mark 10,000 and 10,500. No smoothing is applied.

At update 10,000, mixed L45 sampled success is still 0.91%, while correct attention mass has increased from 2.22% at initialization to 10.08%. By update 10,500, attention mass reaches 47.05% and local correct-transition probability reaches 61.99%, yet sampled terminal success is only 1.11%. Internal changes precede a substantial rise in terminal success.

L45 measurement	Mixed training	Fixed training
Correct attention mass	47.05	3.11
Local correct-transition probability	61.99	1.42
Sampled terminal success	1.11	0.85

Table 2. L45 at update 10,500. All entries are percentages, evaluated on the same 15,360 prompts in both conditions.

The local measurements reveal progress toward solving hard tasks before it becomes visible in terminal success. Under mixed training, attention mass and local transition probability eventually reach about 71.4%, alongside 11.5% sampled terminal success; fixed training ends at only 3.00% attention mass and 1.40% local transition probability. The mixture improves both attention to the intended operations and execution from the current state. These local gains accompany the emergence of successful long-task behavior. Their average does not determine complete-trajectory success, which also depends on where errors occur along each path.

4 Comparisons Based on Reward Gradients

4.1 Reward gradients under mixed and fixed training

At each checkpoint, we measure reward gradients for $L = 5, 15$, and 45 offline, using three fixed 16-prompt batches per length. Measurements occur every 100 training updates and leave Q unchanged. Integrating over generated trajectories gives terminal success probability $P_L(Q; x)$ and its length-normalized gradient:

$$g_L(Q) = \nabla_Q \left[\frac{1}{nL} \sum_{i=1}^n P_L(Q; x_i) \right], \quad n = 16.$$

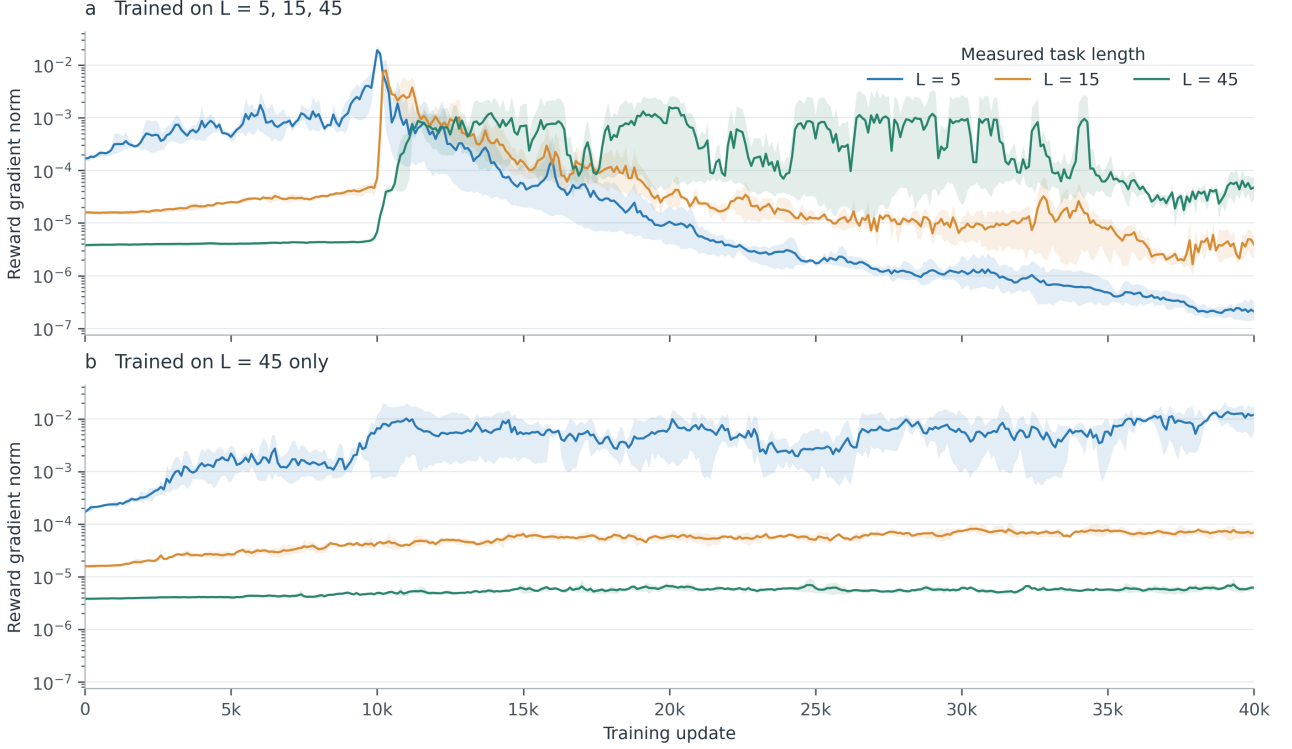


Figure 3. Three reward objectives along each training trajectory. Panel titles specify training lengths; colors specify measurement lengths. Curves within a panel use the same model checkpoints. Lines show mean batch norms; shading spans the three batches. Both panels use identical prompts and axes, $1/L$ normalization, and no smoothing.

Ordered growth and unequal decay. Under mixed training, the $L5$, $L15$, and $L45$ gradients rise in sequence and reach sampled maxima at updates 10,000, 10,300, and 20,000. The temporal resolution is 100 updates, so the exact peak steps may differ. $L5$ and $L15$ then decline with fluctuations; $L45$ remains elevated longer before shifting downward around 35,000. From 17,600 onward, $\|g_{45}\| > \|g_{15}\| > \|g_5\|$ holds at every checkpoint, reversing the initial ordering.

The fixed control. With $L45$ -only training, the measured $L5$ gradient grows to 12.9 times its initial norm by update 5,000 and 39.2 times by 10,000. Yet $\|g_5\| > \|g_{15}\| > \|g_{45}\|$ holds throughout, and the $L45$ norm never exceeds 1.87 times its initial value. Short-task sensitivity can therefore grow while long-task performance remains near chance.

Interpretation. The mixture stays fixed while the relative reward-gradient strengths reverse from short-task to long-task dominance. This is a local signature of an emerging implicit curriculum: the model's changing state reshapes which task lengths supply stronger reward signals. Short-task gradient decay accompanies high short-task success, while increased long-task sensitivity accompanies the emergence of $L45$ success.

The fixed run develops short-task gradients without training on those rewards. In the mixture, shorter tasks supply training examples as the long-task signal grows. This contrast motivates a directional test: can an easier-task ascent step improve $L45$, including after its own gradient has weakened?

4.2 Easy-hard gradient alignment and fine-tuning consistency

At checkpoints of the mixed run, we compare L5-L45 with L15-L45. Gradient alignment measures whether the rewards favor similar parameter changes. A temporary step of fixed size along each easier-task gradient tests *fine-tuning consistency*: whether that direction also improves L45. Three fixed task sets (A-C), each with 16 prompts per length, supply the source gradients and target rewards.

For source length $l = 5$ or 15 , we compute the gradient cosine c and move Q by $\varepsilon = 0.01$ along the unit source gradient. The L45 response Δ measures compatible improvement when positive and interference when negative.

$$c_{l,45} = \frac{\langle g_l, g_{45} \rangle}{\|g_l\| \|g_{45}\|}, \quad Q^+ = Q + \varepsilon \frac{g_l}{\|g_l\|},$$

$$\Delta_{l \rightarrow 45} = \bar{P}_{45}(Q^+) - \bar{P}_{45}(Q).$$

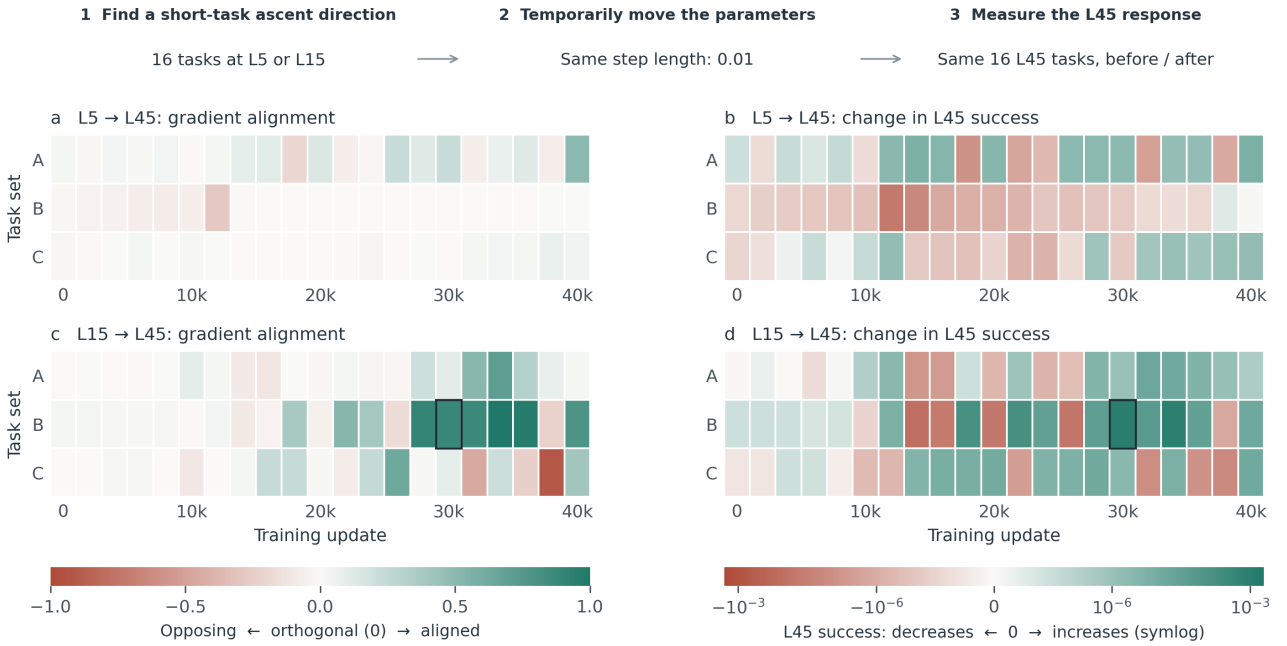


Figure 4. Direction agreement and measured long-task response. Left: gradient cosine; green is aligned, red opposing, and white nearly orthogonal. Right: the fine-tuning response Δ in L45 success probability; green improves success, red reduces it. Each cell is one of three task sets at a checkpoint, every 2,000 updates. L5 and L15 share the L45 target set. Left colors are linear; right colors use a symmetric log scale, linear within $\pm 10^{-8}$. All measurements are shown. Outlines mark the example below.

The largest measured L15 gain occurs for set B at update 30,000 in panels (c,d). The outlined left cell has a cosine of 0.89: the L15 and L45 reward gradients point in similar directions. The matching right cell records $\Delta = 1.11 \times 10^{-3}$, raising L45 success from 8.00% to 8.11%. The paired cells illustrate fine-tuning consistency: an L15 ascent step improves the L45 reward.

Weak early coupling. Through update 10,000, all absolute gradient cosines are below 0.105 and $|\Delta| < 3.35 \times 10^{-7}$. Neither source consistently benefits L45 in this early window.

More frequent late improvement from L15. Over updates 12,000-20,000, L15 steps improve L45 in 8 of 15 measured cells. Over updates 22,000-40,000, they help in 22 of 30 cells, compared with 14 of 30 for L5. L15-L45 alignment and improvement are more consistently positive late in training, with substantial variation across task sets and effect sizes. The counts describe repeated measurements of the same three sets.

Interpretation. Easier-task ascent directions can improve L45. Late in the main run, L15 does so more often even as its gradient weakens. Fixed-size steps isolate directional opportunity from signal strength: useful cross-length directions remain available within the plateau. Their contribution to actual optimizer updates remains to be measured.

5 Computing and validating the local measurements

Expected reward over generated trajectories

To examine the reward signals involved in hard-task learning, we need terminal probabilities even when successful rollouts are rare. For a fixed prompt, the 96-state task permits a finite-state calculation over all generated trajectories. At step k , the frozen MLP defines a transition matrix $T_k(Q)$, whose rows give next-state probabilities. We propagate the initial state distribution and read out the intended endpoint probability:

$$p_0 = e_{s_0}^T, \quad p_k = p_{k-1} T_k, \quad P_L = p_L e_{s_L^*}.$$

A backward recursion gives the derivative with respect to Q . Centering its backward values reduces numerical cancellation: adding a constant to those values contributes zero derivative through a transition row that sums to one. Computation uses float64. This integrates trajectory randomness exactly for the fixed prompts; averaging across prompts still uses the three sampled task sets.

From gradient alignment to a measured response

For the unit source-gradient step in Section 4.2, first-order expansion connects the predicted L45 reward change to its gradient norm and alignment with the source:

$$\Delta_{l \rightarrow 45} \simeq 45 \varepsilon \|g_{45}\| c_{l,45}.$$

The factor 45 reverses the length normalization. Alignment predicts the response sign; the target norm and step size set its first-order magnitude. All 126 responses in Figure 4 have the predicted sign. Positive and negative perturbations at two step sizes also check derivative accuracy (Appendix B). Translating this local opportunity into training progress additionally depends on source gradient strengths, Adam, the baseline, and entropy regularization.

6 Conclusion

Mixed training can bootstrap learning on harder tasks in this independent implementation. It enables success on more long tasks, although final accuracies do not approach one at every length as in the original experiment.

Attention shifts toward the intended operations and local execution improves before terminal success rises appreciably. Reward-gradient norms peak in length order and then decline unevenly. Later in the main run, L15 directions help L45 more often even as the L15 signal weakens. Local opportunities for further improvement thus coexist with a performance plateau.

In seed 101, operation-selection accuracy stops improving while attention sharpens, probe outcomes polarize, and short-task reward gradients shrink (Appendix D). Together, these observations indicate saturation within a partially learned policy despite continued parameter updates. Other seeds reach different accuracy levels; seed 103 retains substantial short-task gradients. Why these learning states arise remains unresolved. The learned transition network and each length's contribution to actual optimizer updates are the next objects of analysis.

AI use

OpenAI Codex assisted extensively with experimental design, code, execution, analysis, figures, and writing. The author directed the study and is responsible for the final report.

Reference

[1] Yu Huang, Zixin Wen, Yuejie Chi, Yuting Wei, Aarti Singh, Yingbin Liang, and Yuxin Chen. *On the Emergence of Implicit Curriculum in RLVR Learning Dynamics*. ICML, PMLR 306:47891-47938, 2026. Conference paper. Experimental exposition: arXiv:2602.14872v3, §7.1, Figures 3-4.

Appendix A Implementation and experimental controls

We independently implement the task and attention architecture in [1], including a pretrained transition head; the authors' code and trained head were not obtained. The table specifies the shared setup and our implementation choices. All formal runs use the same frozen head. Terminal-state reward measures task completion, while the additional full-path metric tests whether every intermediate state follows the intended operation sequence.

Component	Implementation
Task	Additions modulo 96, distinct operations, uniform initial states, and a fixed identifier permutation [1, §§3.1, 7.1]. We use 64 identifiers and task seed 2026.
Attention and head	Class-specific ReLU architecture with operation and state inputs scaled by $1/2$ [1, Eqs. (3)-(4)]. We use 384 units per output class and a 64×64 Q initialized to zero.
Head pretraining	6,000 Adam updates; learning rate 0.003; batch 1,024; seed 17; Gaussian initialization with SD 0.05. Leaky ReLU slope 0.01 during supervision; ReLU during validation and RL.
RL optimizer	Adam at constant learning rate 0.01; betas (0.9, 0.999); epsilon 10^{-8} ; no weight decay.
RL objective	Length-normalized REINFORCE; entropy coefficient magnitude 0.001 [1, §§3.4, 7.1]. In our loss, entropy is averaged over steps and prompts, then subtracted.
Reward baseline	EMA momentum 0.95 [1, §7.1]. We use a global baseline, initially zero. Each update uses the previous baseline, then updates it with the current batch reward.
Training	Batch 512; 40,000 updates; fixed L45 or uniform per-trajectory L5/15/45 mixture. Run seeds 101, 102, and 103. The added five-level mixture uses seed 101.
Evaluation	15,360 prompts per length every 50 updates, including initialization. Greedy and sampled rollouts share prompts; evaluation has an RNG separate from training.
Local measurements	Three fixed sets of 16 prompts per length. Reward gradients every 100 updates; source-to-L45 perturbations every 2,000 updates, with signed steps of 0.01 and 0.005.
Execution	Rented instances with one NVIDIA GeForce RTX 4090; Python 3.12.14 and PyTorch 2.11.0+cu128.

Execution capacity. The frozen head correctly predicts all 9,216 atomic state-operation pairs; the minimum and mean correct probabilities are 0.993072 and 0.999507. Supplying the intended one-hot operation at every step of 8,192 new L45 prompts gives a mean full-path probability of 97.81%. Thus, the head supports long correct trajectories under accurate operation selection. Its response to the soft operation mixtures encountered during learning remains relevant to the observed outcomes.

Matched exposure. Every run processes 20,480,000 trajectories. Within each seed, the fixed and mixed conditions share initialization, head, task permutation, optimizer, and evaluation prompts. Matching updates and trajectories leaves different totals of executed operations.

Fixed-control measurements. The original seed-101 collection omitted fixed-group reward gradients. A rerun with the same seed and configuration reproduced every evaluation metric at all 801 checkpoints. Its gradients supply the fixed panel of Figure 3. The five-level comparison adds measurement lengths using this rerun's saved Q states, without further fixed-group training.

Reproducible evidence. The companion repository contains the report, measured evaluations and local diagnostics, per-update seed-101 records, frozen head, configurations, and implementation. Analysis and plotting commands reconstruct the reported summaries and all eight figures. SHA256 inventories record data and source provenance.

Appendix B Training fluctuations and numerical accuracy

Evaluation success may plateau while the optimizer remains active. Figure 5 records per-update rewards, loss gradients, and parameter changes throughout the main run. These measurements distinguish a stabilized success rate from halted parameter updates.

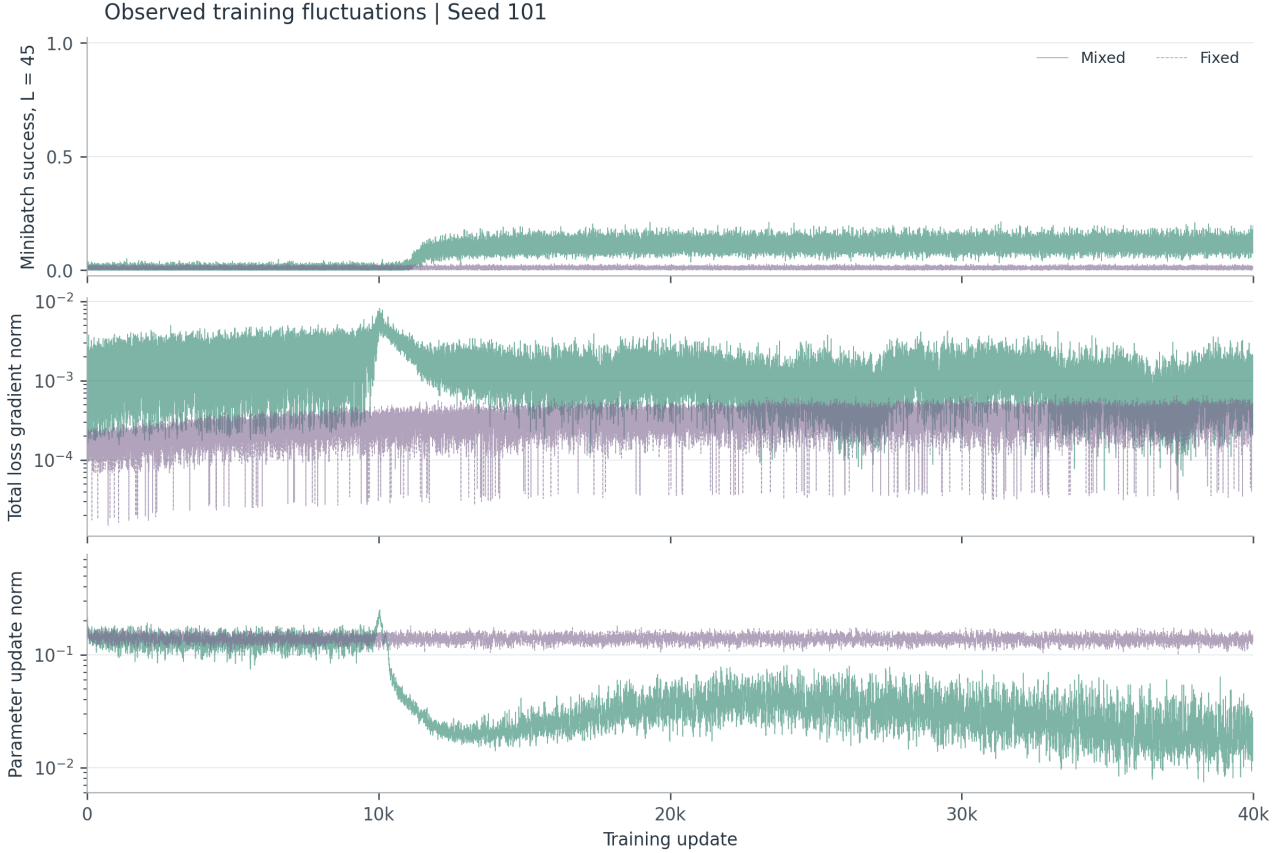


Figure 5. Optimization continues after success rates stabilize. Top: L45 minibatch success; middle: total loss-gradient norm, including baseline and entropy terms; bottom: Adam update norm. All 40,000 updates are shown.

Derivative check. We validate the expected-reward calculation against trajectory enumeration on small tasks, direct differentiation, and independent rollouts. For each source direction $u_l = g_l / \|g_l\|$, a central difference provides a further check at the measured checkpoints:

$$D_\varepsilon = \frac{\bar{P}_{45}(Q + \varepsilon u_l) - \bar{P}_{45}(Q - \varepsilon u_l)}{2\varepsilon}, \quad d = 45 g_{45}^T u_l.$$

Of the 126 source-target cells, 123 agree within 1% at $\varepsilon = 0.01$ and 125 at $\varepsilon = 0.005$. Table 3 gives the three exceptions at the larger step. Their response signs agree with the directional predictions at both sizes, as do those of all other cells.

Update	Source $\rightarrow$ target	Set	Gap at $\varepsilon = 0.01$	Gap at $\varepsilon = 0.005$
8,000	L5 $\rightarrow$ L45	C	1.54%	1.37%
12,000	L15 $\rightarrow$ L45	C	1.18%	0.17%
38,000	L15 $\rightarrow$ L45	A	1.25%	0.90%

Table 3. Relative derivative error. Smaller steps reduce the error; all three cells preserve the response sign.

Relative error is $|D_\varepsilon - d| / \max(|d|, f)$, with cancellation floor $f = 64 u_{\text{mach}} \max(P_+, P_-, P_0) / \varepsilon$; all derivatives exceed this floor. The 1% tolerance tests derivative accuracy. Whether an easier-task step benefits L45 is determined by the signed reward change in Section 4.2. Reusing the prompt sets makes successive checkpoints dependent measurements.

Appendix C Hard-task learning across training seeds

Seeds 102 and 103 repeat both conditions and the full measurement schedule to examine variation in hard-task bootstrapping. Code, hyperparameters, frozen head, and identifier permutation stay fixed. Figure 6 compares how much terminal success develops and how reward-gradient dynamics differ across the three runs.

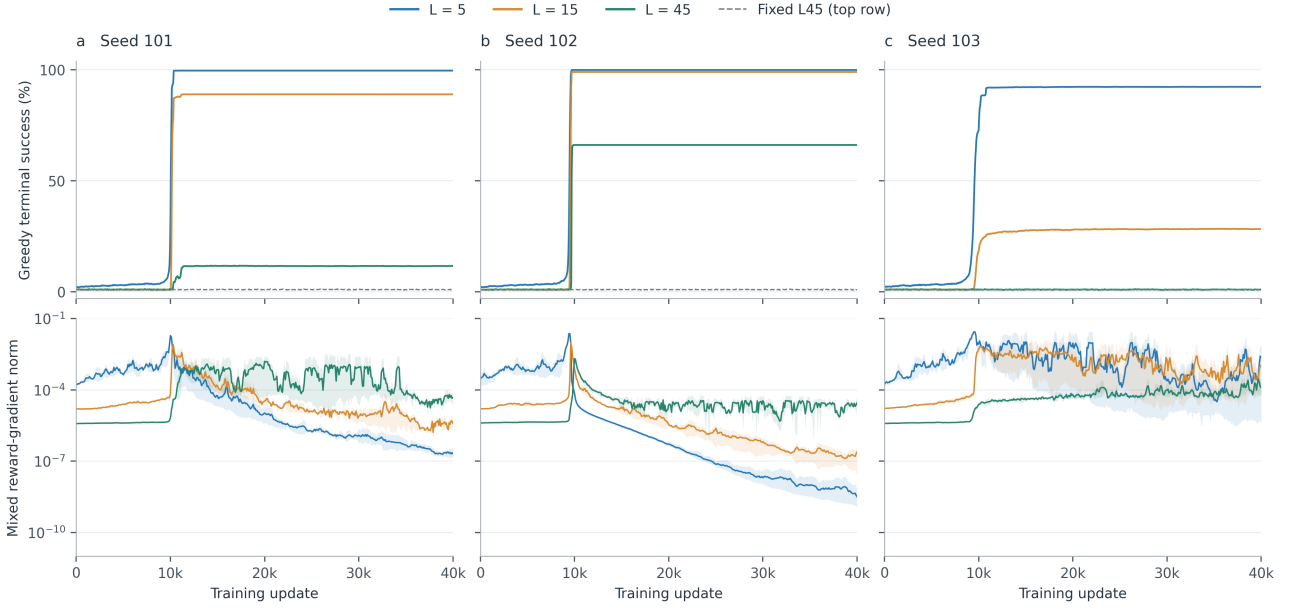


Figure 6. Different outcomes and gradient regimes under one implementation. Each column is a seed. Top: mixed terminal success by length and the dashed fixed-L45 control. Bottom: mean reward-gradient norms, with shading across three probe sets. Axes are shared within rows; all 801 evaluations and 401 gradient checkpoints are retained.

Seed	Metric	Mixed L5	Mixed L15	Mixed L45	Fixed L45
101	Terminal	99.67	88.99	11.65	1.03
101	Full path	91.40	41.11	0.00	0.00
102	Terminal	99.96	99.05	66.17	1.00
102	Full path	97.15	72.58	1.85	0.00
103	Terminal	92.37	28.27	1.01	0.98
103	Full path	66.54	4.02	0.00	0.00

Table 4. Final terminal and full-path success. Percentages on 15,360 prompts per length. Full-path success requires every canonical intermediate state, as well as the correct endpoint.

Different amounts of hard-task progress. Seed 102 reaches 66.17% L45 terminal success, compared with 11.65% in seed 101; seed 103 remains near chance. The same mixture can therefore enable successful outcomes on very different fractions of long tasks. Seed 102’s full-path success is 1.85%, showing that many correct endpoints still arise through noncanonical intermediate states.

Different local regimes. Seed 102 repeats the ordered gradient peaks, at updates 9,400, 9,600, and 10,000, followed by strong short-task decay. Seed 103 retains substantial L5 and L15 gradients. Its L45 attention mass and local transition probability reach 41.75% and 41.68%, despite chance-level terminal success. Internal progress, usable reward signals, and terminal performance can therefore separate in different ways across runs.

Choice of the main case. Seed 101 is the original run. Before the additional runs, we set a replacement criterion of at least 99% greedy terminal success at every trained length throughout updates 39,000-40,000. Neither new run reaches that accuracy threshold, so seed 101 remains the main example. All three outcomes are reported.

Appendix D Operation selection in the L15 plateau

Peak attention-hit rate, also measured in [1, Figure 3b], counts steps at which the largest attention weight selects the intended operation. We compare it with the attention mass from Section 3 to distinguish a gain in selection accuracy from increasing weight on an existing selection. Figure 7 follows this discrete accuracy under fixed and mixed training.

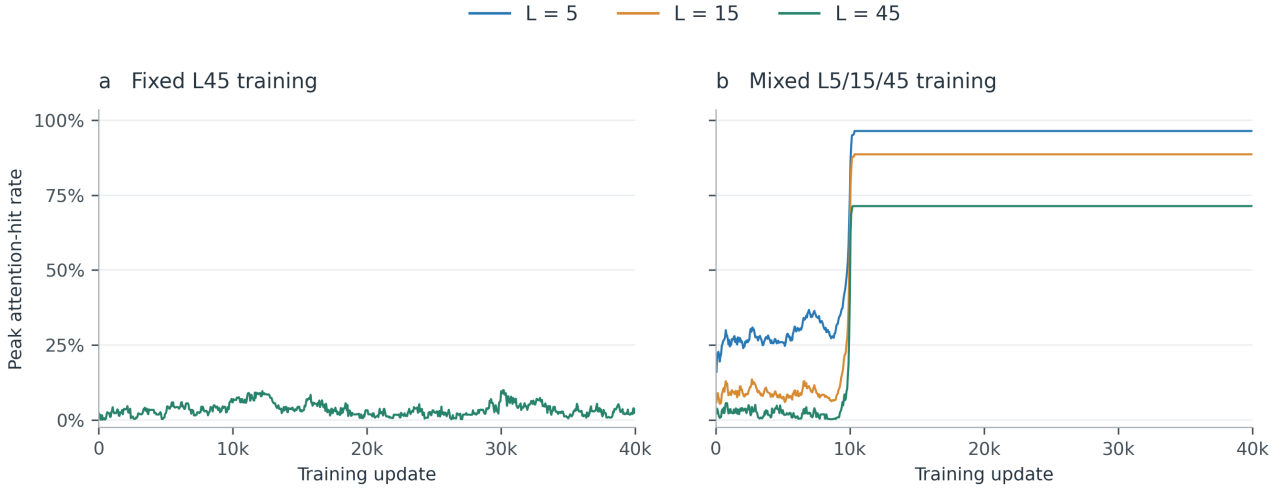


Figure 7. Accuracy of operation selection. The left panel follows the fixed-L45 run; the right follows mixed L5/15/45 training. Colors identify evaluation lengths. Both panels use the same 801-checkpoint grid as Figure 1. The mixed-condition measurement extends the source paper’s fixed-length attention comparison.

Condition	Attention hit	Full path
Fixed, L = 45	2.74	0.00
Mixed, L = 5	96.45	91.40
Mixed, L = 15	88.66	41.11
Mixed, L = 45	71.38	0.00

Table 5. Operation selection and complete trajectories at update 40,000. Percentages on 15,360 prompts. Attention hit averages position selection over steps; full-path success requires every canonical state along the trajectory.

Sharpening within a partial solution. L15 full-path success stays at 41.11% from update 10,400. Between updates 12,000 and 40,000, correct attention mass rises from 83.08% to 88.67%, while aggregate attention-hit rate remains 88.66%. The allocation sharpens without a net gain in correct position selection. Meanwhile, the L15 reward-gradient norm falls about 2,054-fold from its peak, even as parameter updates continue.

Polarized prompt outcomes. At the final checkpoint, 42 of the 48 fixed L15 probe prompts average 99.26% terminal success probability; the remaining six average 0.029%. These groups are defined retrospectively from the probe set. This separation, together with sharpening attention and diminishing gradients, characterizes saturation within a partially learned policy even as optimization continues.

What terminal reward reinforces. Among rewarded L15 training trajectories during updates 20,001-40,000, 54.38% contain an incorrect intermediate state. They receive the same terminal reward as fully correct paths. Learning can therefore produce more successful endpoints without acquiring faithful execution of the entire operation sequence. The learned head’s behavior under soft operation mixtures is a natural target for further analysis.

Appendix E A five-level mixture retaining L45

We add a mixture of L5, L10, L20, L40, and L45 to examine denser intermediate coverage while keeping the same longest task. Lengths are sampled uniformly from the first update. The schedule doubles from 5 until the cap at 45; L15 is measured but never trained. This is a five-level experiment. The three-level L5/10/20 design remains untested.

Seed 101, the frozen head, optimizer, batch size, and update horizon match the main run. Changing the mixture reduces the L45 sampling share from one third to one fifth and increases mean task length from 21.67 to 24. The comparison measures this joint change in task composition and exposure.

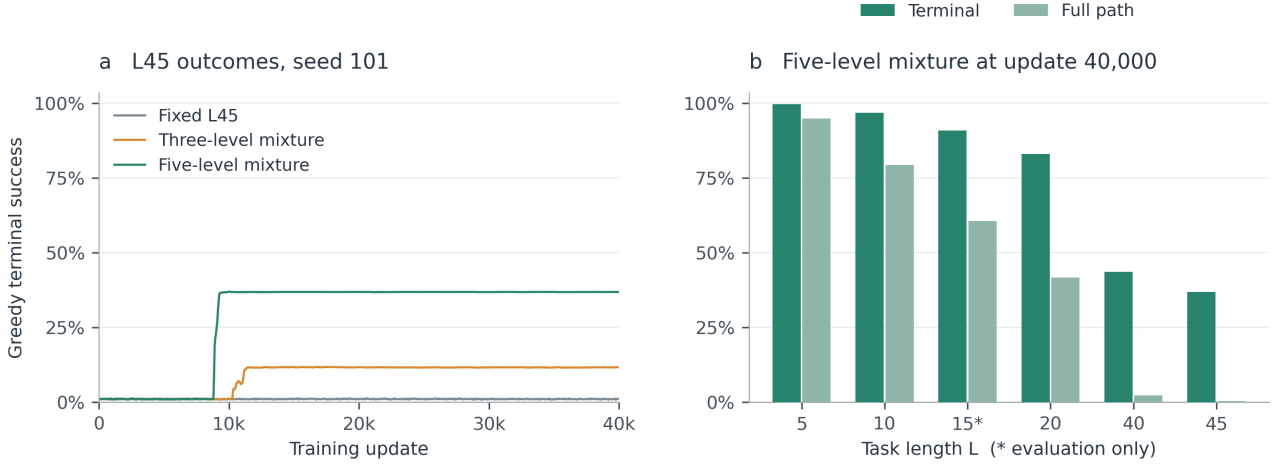


Figure 8. More long-task success with a five-level mixture. (a) L45 terminal success for fixed training and the three- and five-level mixtures, all with seed 101. (b) Terminal and full-path success of the five-level run at update 40,000. Each rate uses 15,360 prompts; the asterisk marks untrained L15. All 801 time points are retained.

Bootstrapping a larger fraction of long tasks. L45 terminal success reaches 36.88%, compared with 11.65% in the three-level seed-101 run and 1.03% under fixed training. L45 full-path success is 0.39%. The terminal rate later stays between 36.82% and 36.88% over updates 20,000-40,000. The changed mixture enables more long-task success in this seed, while three-level seed 102 reaches a higher 66.17%.

Ordered sensitivity, uneven coupling. Mean reward-gradient norms peak at updates 8,600, 8,700, 8,800, 9,000, 9,200, and 9,200 for L5, L10, L15, L20, L40, and L45. The two longest lengths share a peak on the 100-update grid. At the final checkpoint, norms increase with length. At $\epsilon = 0.01$, L40 ascent directions help L45 in 57 of 63 cells, whereas L20 directions help it in 25. All 315 responses have the predicted sign. This run also combines hard-task progress with remaining opportunities for local cross-length improvement.

Matched measurements. All six lengths use three fixed sets of 16 prompts and 100-update gradient measurements. Every 2,000 updates, we step along each shorter-length reward gradient at $\epsilon = 0.01$ and 0.005 and measure the L45 response. Fixed-group gradients are computed from its 801 saved Q states; all 3,609 measurements at the original lengths match the earlier values exactly.